

VECTOR — Current Research State

Last synced: 2026-09-24
Scope: Concise research dashboard; detailed evidence and history remain in the linked sources. This is an initial state record, not a new empirical verification.

Current scientific question

How do assortative assignment, congestion, and selection scarcity K/N interact? Does assignment preference create, amplify, or reshape congestion's effect on advancement? Can congestion change outcomes without assortative preference, and could extreme scarcity make men's basketball (MBB) a boundary regime? Similar outcome curves may arise from different mechanisms. These are open questions. [S1–S2]

Current advisor priorities

The latest advisor discussion represented in the reviewed brief is PD41 (September 22): resolve whether assortativity matters before reshaping the presentation. PD40 explicitly allows the answer that assortativity is unnecessary and cautions against generalizing from basketball alone. This dashboard relies on the brief's transcript review; it does not establish that no later advisor discussion exists. Charles reaffirmed this focus on September 24. [S1; S2 §1]

Established evidence and present baseline

- **Binding framework:** distinguish ASSIGN, SCORE, and SELECT; an observed outcome curve does not identify its mechanism. The binding document separately distinguishes peer environment from advancement. [S2 §2; S3]
- **Directly inspected documentation:** the working brief specifies an eight-cell comparison crossing ρ , λ , and K/N . The design exists; this experiment has not been executed. [S2 §§3–4]
- **Saved, unreproduced output:** the brief records a Grandchild sorting sweep with positive realized sorting at $\rho = 0$. This supports keeping assignment preference separate from realized sorting; it is not a result for the proposed factorial population. [S2 §5, S11 therein]
- **Army status:** the September 24 handoff reports Runs 1–3 completed, Runs 4–5 pending, and a planned own-zero plot gate. These are reported progress, not independently verified refreshed figures; the refresh remains **PARTIAL**. [S2 §6b; S4]

No additional reproduced empirical finding is established by this dashboard.

Active hypotheses

Extreme scarcity may permit congestion effects without assortative preference. Assortativity may instead amplify or reshape effects, and different parameter combinations may produce similar curves. Army's band-of-excellence framing is a complementary mechanism hypothesis. These hypotheses are under design and evidence review, not experimental testing. [S1–S2]

Unresolved contradictions

- **Basketball:** calibration versus replay probabilities and parameter transfer; annual versus ever-drafted outcomes; empirical versus simulated axes; local tail decline versus global quadratic curvature.
- **Army:** straight-LOO versus earlier peer definitions; $H_{\text{sort}} = 0.131$ versus approximately 0.255–0.279; Mac/AWS run-to-result mapping; completed components versus unfinished mosaic.
- **Tenure:** construction roster versus inference sample versus decision cohort; person-year versus decision-year analyses; completed versus unfinished survival work.
- **Terminology:** historical Pass A/B/C and "Layer B" labels must retain their document-specific meanings.

These are carried-forward reconciliation questions, not newly verified findings or discrepancies resolved by this update. [S2 §7; prior reconciliation questions retained by Charles]

Current task

Maintain the PD41 working brief and evidence map in scientific-design and evidence-analysis mode. Preserve the calibration/replay limitation without letting it displace the immediate assortativity investigation. [S1]

Decisions made

Charles authorized routine maintenance of VECTOR-owned research documents in this workspace and designated the brief a working document. Substantive scientific changes must be explained before adoption as authoritative. Analytical execution and Git operations require separate explicit authorization. The source operating rules govern the full scope. [S1]

Immediate next decision/action

Discuss the brief's proposed exact, unclipped top- K rule; fixed viability and scaling assumptions; scenario values; tolerances; and paired replication plan. Before requesting execution, specify the exact files, held-fixed quantities, manipulations, outcome interpretations, and provenance risks. No execution approval is recorded. [S1 §7; S2 §8]

Sources

- **S1:** [Charles's standing repository authority](#).
- **S2:** [PD41 working scientific brief](#).
- **S3:** [Binding separation of environment, scoring, and selection](#).

- **S4:** [September 24 Army handoff.](#)